

GATE- ALL BRANCHES ENGINEERING MATHEMATICS



Linear Algebra

Lecture No. 6



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[Recap of previous lecture]



→ Vector Spaces

↳ Basis

↳ $\dim(V)$

↳ $\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$

$$\rightarrow A_{m \times n} \vec{x}_{n \times 1} = \vec{b}_{m \times 1}$$

Consistency.

$$\rho(A) = \rho(A|\vec{b}) = n \rightarrow \text{unique solution}$$

$$\rho(A) = \rho(A|\vec{b}) < n \rightarrow \text{Infinitely Many solutions}$$

$$\rho(A) \neq \rho(A|\vec{b}) \rightarrow \text{No solution.}$$

Topics to be Covered



Topic

✓ System of Linear Equations (contd.)

Topic

✓ Eigen Values and Eigen Vectors



System of Linear Equations (Contd.)

consider a system $A_{m \times n} \vec{x}_{n \times 1} = \vec{b}_{n \times 1}$; say $\rho(A) = \rho(A | \vec{b}) < n$.
i.e the system has Infinitely Many solutions.

Let $\vec{x}_1$ & $\vec{x}_2$ be two solutions of $A\vec{x} = \vec{b}$
 $\Rightarrow A\vec{x}_1 = \vec{b}$ & $A\vec{x}_2 = \vec{b}$.

Since systems are linear,

By superposition, Any linear combination of $\vec{x}_1$ & $\vec{x}_2$ is also a solution.
i.e $(\alpha_1 \vec{x}_1 + \alpha_2 \vec{x}_2)$ is also a solution.

→ For a system $A_{m \times n} \cdot \vec{x}_{n \times 1} = \vec{b}_{m \times 1}$ having infinitely Many solutions, the Number of linearly independent solutions is $n - \rho(A)$.

Let the solutions be $\boxed{\vec{x}_1, \vec{x}_2}, \vec{x}_3, \dots$

For Ex: If $n - \rho(A) = 2$; then

$\vec{x}_3, \vec{x}_4, \vec{x}_5, \dots$ are some linear combinations of $\vec{x}_1$ & $\vec{x}_2$.

→ In general Any linear Equation is $\underbrace{a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n}_{\text{Degree} = 1} = \underbrace{b_1}_{\text{Constant (zero degree term)}}$

Non-Homogeneous Equation.

No Variable

→ System of Homogeneous Equations:

$$A_{m \times n} \cdot \vec{x}_{n \times 1} = \vec{0}_{m \times 1}$$

For this system, $\vec{x} = \vec{0}$ is always a solution;

It is called Trivial Solution.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = 0 \\ a_{21}x_1 + a_{22}x_2 = 0. \end{cases}$$

$$x_1 = 0; x_2 = 0$$

→ Consider $[A | \vec{b}]$; If $\rho(A) = \rho(A | \vec{b}) < n$;

then the system has infinitely Many solutions.

Then We say $\exists$ Non-Trivial Solution for the homogeneous system.

→ For a homogeneous system;

$$A_{n \times n} \vec{x}_{n \times 1} = \vec{0}_{n \times 1}$$

Consider $[A|\vec{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & 0 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} & 0 \end{array} \right]_{n \times (n+1)}$

Max. Possible Minors of $[A|\vec{b}]$ are of size $(n \times n)$.

$$\rho(A|\vec{b}) = \rho(A)$$

★ $\therefore$ For $\rho(A)$ being ' n ' or ' $< n$ ' depends on $\det(A)$.

For $A_{n \times n} \vec{x}_{n \times 1} = \vec{0}_{n \times 1}$ to have Non-Trivial solution;

$$\rho(A_{n \times n}) < n \Rightarrow \boxed{\det(A) = 0}$$

Ex: Consider

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= 0. \end{aligned}$$

$$\Rightarrow [A|\vec{b}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \end{bmatrix}_{3 \times 4}$$

3x3 Minors:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}; \begin{vmatrix} a_{12} & a_{13} & 0 \\ a_{22} & a_{23} & 0 \\ a_{32} & a_{33} & 0 \end{vmatrix}; \begin{vmatrix} a_{11} & a_{13} & 0 \\ a_{21} & a_{23} & 0 \\ a_{31} & a_{33} & 0 \end{vmatrix}; \begin{vmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & 0 \end{vmatrix}$$

$$\Rightarrow \mathcal{P}(A|\vec{b}) \text{ depends on } \det(A)$$

→ Solving Techniques for system of Equations: (Direct Methods).



① Gauss Elimination:

For a linear system of equations $A\vec{x} = \vec{b}$.

- a) Convert $[A|\vec{b}]$ to Row echelon form.
- b) Do backward substitution to get the vector $\vec{x}$.

(OR).

- a) Perform Backward elimination.
- b) Do forward substitution.

→ Consider a 3×3 System,

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Form $[A|\vec{b}] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$

$$R_2 \rightarrow R_2 - \left(\frac{a_{21}}{a_{11}}\right)R_1$$

$$R_3 \rightarrow R_3 - \left(\frac{a_{31}}{a_{11}}\right)R_1$$

$$[A|\vec{b}] \sim \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & a'_{32} & a'_{33} & b'_3 \end{array} \right]$$

$$R_3 \rightarrow R_3 - \left(\frac{a'_{32}}{a'_{22}}\right)R_2$$

$$[A|\vec{b}] \sim \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a''_{33} & b''_3 \end{array} \right]$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2$$

$$a''_{33}x_3 = b''_3$$

By Backward substitution, x_3, x_2, x_1 can be calculated.

→ Say we have $A\vec{x} = \vec{b}$.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow [A|\vec{b}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$$

By Row transformations.

$$[A|\vec{b}] \sim \begin{bmatrix} a'_{11} & 0 & 0 & b'_1 \\ a'_{21} & a'_{22} & 0 & b'_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$$

The system converts to

$$\Rightarrow a'_{11}x_1 = b'_1; \quad a'_{21}x_1 + a'_{22}x_2 = b'_2; \quad a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3.$$

→ solve

$$\begin{aligned} x+y+z &= 6 \\ 2x-y+z &= 3 \\ x-y+2z &= 5 \end{aligned}$$

Sol:

$$[A|\vec{b}] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & -1 & 2 & 5 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & -2 & 1 & -1 \end{array} \right]$$

$$R_3 \rightarrow 3R_3 - 2R_2$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 0 & 5 & 15 \end{array} \right]$$

$$\begin{aligned} \Rightarrow x + y + z &= 6 \Rightarrow x = 1 \\ -3y - z &= -9 \Rightarrow y = 2 \\ 5z &= 15 \Rightarrow z = 3 \end{aligned}$$

$$\Rightarrow \vec{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

② LU Decomposition: $\rightarrow$ use this Method only for
'n' Equations with 'n' unknowns.

Consider $A\vec{x} = \vec{b}$.

We decompose A as Product to a lower Triangular & an upper Triangular Matrix.

$\boxed{A = LU}$ Where $L \rightarrow$ Lower Triangular Matrix.
 $U \rightarrow$ Upper Triangular Matrix.

$\therefore A\vec{x} = \vec{b}$
 $\Rightarrow L(U\vec{x}) = \vec{b}$

Let $U\vec{x} = \vec{y} \Rightarrow$ The system is $L\vec{y} = \vec{b}$.

$$\begin{bmatrix} - & 0 & 0 \\ - & - & 0 \\ - & - & - \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

By Forward Substitution, We can calculate ' $\vec{y}$ '.

Since $U\vec{x} = \vec{y} \Rightarrow$ By Backward Substitution, We get $\vec{x}$. ✓

→ Factorising 'A' as LU:

a) Doolittle Algorithm:

$A = LU$ Where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}_{3 \times 3} \quad \& \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}_{3 \times 3}$$

$$A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

b) Croute's Algorithm:

$$A = LU$$

$$\underline{L} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}; \underline{U} = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

Ex. → Consider the system.

$$\left. \begin{array}{l} x+y+z=6. \\ 2x-y+z=3 \\ x-y+2z=5 \end{array} \right\} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & -1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 6 \\ 3 \\ 5 \end{bmatrix}}_{\vec{b}}.$$

Consider $A=LU$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \cdot \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_{13} \\ u_{23} \\ u_{33} \end{bmatrix}$$

$$\Rightarrow \boxed{u_{11}=1; u_{12}=1; u_{13}=1}$$

$$\begin{array}{l} 2 = l_{21} \cancel{u_{11}}^1 \\ \Rightarrow \boxed{l_{21}=2} \end{array} \quad \begin{array}{l} -1 = \cancel{l_{21}}^2 \cancel{u_{12}}^1 + u_{22} \\ \Rightarrow \boxed{u_{22}=-3} \end{array} \quad \begin{array}{l} 1 = \cancel{l_{21}}^2 \cancel{u_{13}}^1 + 1 \cdot u_{23} \\ \Rightarrow \boxed{u_{23}=-1} \end{array}$$

$$1 = l_{31} \cancel{u_{11}}^1 \Rightarrow \boxed{l_{31}=1}$$

$$-1 = \cancel{l_{31}}^1 \cancel{u_{12}}^1 + l_{32} \cancel{u_{22}}^{-3}$$

$$\Rightarrow \boxed{l_{32}=2/3}$$

$$2 = \cancel{l_{31}}^1 \cancel{u_{13}}^1 + \cancel{l_{32}}^{2/3} \cancel{u_{23}}^{(-1)} + u_{33}$$

$$\Rightarrow \boxed{u_{33}=5/3}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & 5/3 \end{bmatrix}$$

$$\underbrace{A = LU}$$

$$\text{Let } U\vec{x} = \vec{y} \Rightarrow L\vec{y} = \vec{b}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2/3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 5 \end{bmatrix}$$

$$\Rightarrow y_1 = 6; \quad 2y_1 + y_2 = 3 \Rightarrow y_2 = -9$$

$$y_1 + \frac{2}{3}y_2 + y_3 = 5 \Rightarrow y_3 = 5$$

$$\vec{y} = \begin{pmatrix} 6 \\ -9 \\ 5 \end{pmatrix}$$

$$\text{since } U\vec{x} = \vec{y}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & 5/3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -9 \\ 5 \end{bmatrix}$$

$$\Rightarrow \frac{5}{3}z = 5, \quad -3y - z = -9; \quad x + y + z = 6$$

$$\Rightarrow z = 3, \quad y = 2, \quad x = 1$$

$$\Rightarrow \vec{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

The value of α for which the system of equations

$$x - y - 3z = 3$$

$$2x + z = 0$$

$$-2y - 7z = \alpha$$

Has a solution is 6.

80: Given system is

$$\underbrace{\begin{bmatrix} 1 & -1 & -3 \\ 2 & 0 & 1 \\ 0 & -2 & -7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 3 \\ 0 \\ \alpha \end{bmatrix}}_{\vec{b}}$$

$$\Rightarrow \alpha - 6 = 0$$

$$\Rightarrow \alpha = 6$$

Consider: $[A|\vec{b}] = \begin{bmatrix} 1 & -1 & -3 & 3 \\ 2 & 0 & 1 & 0 \\ 0 & -2 & -7 & \alpha \end{bmatrix}$

$R_2 \rightarrow R_2 - 2R_1$

$$[A|\vec{b}] \sim \begin{bmatrix} 1 & -1 & -3 & 3 \\ 0 & 2 & 7 & -6 \\ 0 & -2 & -7 & \alpha \end{bmatrix}$$

$R_3 \rightarrow R_3 + R_2$

$$\begin{bmatrix} 1 & -1 & -3 & 3 \\ 0 & 2 & 7 & -6 \\ 0 & 0 & 0 & \alpha - 6 \end{bmatrix}$$

For solution to exist;

$$\Rightarrow \rho(A) = 2; \rho(A|\vec{b}) = 2$$

MCQ

Consider the system of linear equations

$$2x_2 + x_3 = 0$$

$$-2x_1 - x_3 = 0$$

$$-x_1 + x_2 = 1$$

The above system has

Sol: Given system is

$$\begin{bmatrix} -2 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow [A|\vec{b}] = \begin{bmatrix} -2 & 0 & -1 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow 2R_2 - R_1} [A|\vec{b}] \sim \begin{bmatrix} -2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} [A|\vec{b}] \sim \begin{bmatrix} -2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$\boxed{\rho(A|\vec{b}) = 3}$$

$$\rho(A) = 2 \Rightarrow \rho(A) \neq \rho(A|\vec{b})$$

- ☐ A a unique solution
- ☐ B infinite number of solutions
- ☒ C no solution
- ☐ D only two distinct solutions

$$\begin{aligned}
 & \left. \begin{aligned} 2x_2 + \cancel{x_3} &= 0 \\ -2x_1 - \cancel{x_3} &= 0 \end{aligned} \right\} \rightarrow \text{on Adding} & -2x_1 + 2x_2 &= 0 \\
 & \boxed{-x_1 + x_2 = 1} & \Rightarrow & \boxed{-x_1 + x_2 = 0}
 \end{aligned}$$

No x_1, x_2
 Can satisfy
 these two simultaneously.

Let r and s be real numbers. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ r & s & 0 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ s-1 \end{pmatrix}$, then the system of linear equations $AX = b$ has

- ☐ A no solutions for $s \neq 2r$.
- ☐ B infinitely many solutions for $s = 2r \neq 2$.
- ☐ C a unique solution for $s = 2r = 2$.
- ☐ D infinitely many solutions for $s = 2r = 2$.



2 mins Summary



- system (Homogeneous).
- linearly Independent solutions.
- Gauss Elimination
- LU Decomposition.



THANK - YOU